CHEM 1211K Exam 2 — Answer Key with Worked Solutions

Problem numbers match the practice set (e.g., 7B = the second new problem for original Question 7). Final answers are in bold. Try every problem before you check here.

Part I — Solutions and Stoichiometry

Original Question 1 · Dilution

1A. 25.0 mL of 12.0 M HCl is diluted to a final volume of 0.500 L. What is the molarity of the diluted acid?

Solution: $M_2 = M_1V_1/V_2 = (12.0\ M)(0.0250\ L)/(0.500\ L) =$ **0.600 M**

1B. A student adds 150.0 mL of water to 50.0 mL of 1.80 M KNO_3. Assuming volumes are additive, what is the new concentration? (Watch out: 150.0 mL is not the final volume.)

Solution: $V_2 = 50.0 + 150.0 = 200.0$ mL. $M_2 = (1.80)(50.0)/(200.0) =$ **0.450 M**

1C. Serial dilution: 10.0 mL of 1.00 M $CuSO_4$ is diluted to 100.0 mL (solution A). Then 5.00 mL of solution A is diluted to 250.0 mL (solution B). What is the molarity of solution B?

Solution: A: (1.00)(10.0)/100.0 = 0.100 M. B: (0.100)(5.00)/250.0 = **2.00×10^{-3} M**

Original Question 2 · Gas stoichiometry with a limiting reactant

2A. $CO_2(g) + 2\ NaOH(aq) \rightarrow Na_2CO_3(aq) + H_2O(l)$
What volume (L) of CO_2 at 25.0 °C and 0.980 atm is needed to react completely with 0.250 L of 1.20 M NaOH?

Solution: mol NaOH = (1.20)(0.250) = 0.300 mol → mol CO_2 = 0.300/2 = 0.150 mol.
$V = nRT/P = (0.150)(0.08206)(298.15)/0.980 =$ **3.74 L**

2B. $Zn(s) + 2\ HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$
3.27 g of Zn is added to 50.0 mL of 1.50 M HCl. (a) Which reactant is limiting? (b) What volume of H_2 forms at STP (0 °C, 1.00 atm)? (c) How many moles of the excess reactant remain?

Solution: mol Zn = 3.27/65.38 = 0.0500 mol; mol HCl = (1.50)(0.0500) = 0.0750 mol.
0.0750 mol HCl needs only 0.0375 mol Zn, so **(a) HCl is limiting**.
(b) mol H_2 = 0.0750/2 = 0.0375 mol; V = (0.0375)(22.41 L/mol) = **0.841 L**
(c) Zn left = 0.0500 − 0.0375 = **0.0125 mol Zn**

Original Question 3 · Ion concentrations after mixing (common ion)

3A. 40.0 mL of 0.500 M K_2SO_4 is mixed with 60.0 mL of 0.300 M KCl (total volume 100.0 mL). Find $[K^+]$, $[SO_4^{2-}]$, and $[Cl^-]$.

Solution: K_2SO_4: 0.0200 mol → 0.0400 mol K^+, 0.0200 mol SO_4^{2-}. KCl: 0.0180 mol → 0.0180 mol K^+, 0.0180 mol Cl^-.
$[K^+] = 0.0580/0.1000 =$ **0.580 M**; $[SO_4^{2-}] =$ **0.200 M**; $[Cl^-] =$ **0.180 M**

3B. How many grams of $MgCl_2$ (95.21 g/mol) must be dissolved in 250.0 mL of 0.100 M NaCl so that the final $[Cl^-]$ is 0.500 M? Assume the volume does not change.

Solution: Cl^- needed = (0.500)(0.2500) = 0.125 mol; Cl^- already present = 0.0250 mol; need 0.100 mol more.
Each $MgCl_2$ gives 2 Cl^-: 0.0500 mol $MgCl_2$ × 95.21 = **4.76 g**

Original Question 4 · Molarity and ions from a formula

4A. A solution of $Fe_2(SO_4)_3$ has $[SO_4^{2-}]$ = 0.450 M. (a) What is $[Fe_2(SO_4)_3]$? (b) What is $[Fe^{3+}]$? (c) What mass of $Fe_2(SO_4)_3$ (399.88 g/mol) is in 500.0 mL of this solution?

Solution: (a) 0.450/3 = **0.150 M** (b) 2 × 0.150 = **0.300 M Fe^{3+}**
(c) (0.150)(0.5000) = 0.0750 mol × 399.88 = **30.0 g**

4B. Which solution has the highest total ion concentration? 0.10 M Na_3PO_4, 0.15 M $CaCl_2$, 0.20 M KNO_3, or 0.10 M $Al_2(SO_4)_3$.

Solution: Na_3PO_4: 4 ions × 0.10 = 0.40 M. $CaCl_2$: 3 × 0.15 = 0.45 M. KNO_3: 2 × 0.20 = 0.40 M. $Al_2(SO_4)_3$: 5 × 0.10 = 0.50 M.
0.10 M $Al_2(SO_4)_3$ is highest.

Part II — Thermochemistry

Original Question 5 · Open, closed, and isolated systems

5A. Classify each: (a) a sealed can of soda cooling in a refrigerator (b) a human body (c) a candle burning inside a sealed glass jar (d) a perfectly insulated, sealed thermos (e) a pot of soup with the lid off on a stove.

Solution: (a) **closed** (heat leaves, no matter) (b) **open** (c) **closed** (d) **isolated** (e) **open** (steam escapes)

5B. A pot of water boils on a stove with its lid off. What single change would turn it into a closed system? What would make it (ideally) isolated?

Solution: Seal it with a lid that does not let vapor out → **closed** (heat still flows in). Seal it and perfectly insulate it from the stove and air → **isolated**.

Original Question 6 · Signs of q, w, and ΔU

6A. A gas in a piston absorbs 450 J of heat while expanding from 2.00 L to 5.50 L against a constant external pressure of 1.20 atm. Find q, w (in J), and ΔU. (1 L·atm = 101.3 J)

Solution: q = **+450 J**. w = −PΔV = −(1.20)(3.50) = −4.20 L·atm × 101.3 = **−426 J**.
ΔU = 450 + (−426) = **+24 J**

6B. The surroundings compress a gas, doing 325 J of work on it, while the gas releases 510 J of heat. Give the signs and values of q, w, and ΔU.

Solution: q = **−510 J**, w = **+325 J**, ΔU = **−185 J**

6C. An exothermic reaction happens in a sealed, rigid steel container. What are the signs of q, w, and ΔU?

Solution: Rigid → ΔV = 0 → **w = 0**. Exothermic → **q < 0**. ΔU = q → **ΔU < 0**.

Original Question 7 · Hess's law

7A. Use these equations to find ΔH for $C(s) + \frac{1}{2} O_2(g) \rightarrow CO(g)$:
(1) $C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H = -393.5$ kJ
(2) $CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$ $\Delta H = -283.0$ kJ

Solution: Keep (1), reverse (2): $CO_2 \rightarrow CO + \frac{1}{2} O_2$, $\Delta H = +283.0$ kJ.
Sum: $\Delta H = -393.5 + 283.0 =$ **-110.5 kJ**

7B. Given:
(1) $N_2(g) + O_2(g) \rightarrow 2\ NO(g)$ $\Delta H = +180.6$ kJ
(2) $2\ NO(g) + O_2(g) \rightarrow 2\ NO_2(g)$ $\Delta H = -114.2$ kJ
Find ΔH for $NO_2(g) \rightarrow \frac{1}{2} N_2(g) + O_2(g)$.

Solution: Add (1) + (2): $N_2 + 2\ O_2 \rightarrow 2\ NO_2$, $\Delta H = +66.4$ kJ.
Reverse: $2\ NO_2 \rightarrow N_2 + 2\ O_2$, $\Delta H = -66.4$ kJ. Halve: **$\Delta H = -33.2$ kJ**

Original Question 8 · Coffee-cup calorimetry (neutralization)

8A. 50.0 mL of 1.00 M HCl and 75.0 mL of 0.800 M NaOH, both at 22.15 °C, are mixed in a calorimeter with $C_{cal} = 18.0$ J/°C. $\Delta H_{neut} = -55.8$ kJ per mol H_2O. Density = 1.00 g/mL, c = 4.184 J/(g·°C). What is the final temperature?

Solution: mol HCl = 0.0500, mol NaOH = 0.0600 → HCl limiting → 0.0500 mol H_2O.
Heat released = (0.0500)(55.8 kJ) = 2.79 kJ = 2790 J.
Total heat capacity = (125.0 g)(4.184) + 18.0 = 541.0 J/°C. ΔT = 2790/541.0 = 5.16 °C.
$T_f = 22.15 + 5.16 =$ **27.31 °C**

8B. To find a calorimeter's heat capacity, 50.0 g of water at 60.0 °C is poured into 50.0 g of water at 20.0 °C already inside the calorimeter (also at 20.0 °C). The final temperature is 39.2 °C. What is C_{cal}?

Solution: Heat lost by hot water = (50.0)(4.184)(20.8) = 4351 J. Heat gained by cold water = (50.0)(4.184)(19.2) = 4017 J.
Calorimeter gained 334 J over ΔT = 19.2 °C → $C_{cal} =$ **17.4 J/°C**

Original Question 9 · ΔH°rxn from enthalpies of formation

9A. $C_2H_5OH(l) + 3\ O_2(g) \rightarrow 2\ CO_2(g) + 3\ H_2O(l)$ $\Delta H°rxn = -1366.8$ kJ
Given $\Delta H°_f[CO_2(g)] = -393.5$ kJ/mol and $\Delta H°_f[H_2O(l)] = -285.8$ kJ/mol, find $\Delta H°_f$ of ethanol.

Solution: Products: 2(−393.5) + 3(−285.8) = −1644.4 kJ.
$-1366.8 = -1644.4 - [\Delta H°_f(C_2H_5OH) + 3(0)] \rightarrow \Delta H°_f =$ **-277.6 kJ/mol**

9B. Which of these have $\Delta H°_f = 0$? $O_2(g)$, $O_3(g)$, C(graphite), C(diamond), $Br_2(l)$, $Br_2(g)$, Na(s), H(g)

Solution: **$O_2(g)$, C(graphite), $Br_2(l)$, Na(s)**: these are the elements in their most stable form at 25 °C and 1 bar. O_3, diamond, $Br_2(g)$, and atomic H are not.

Original Question 10 · Specific heat: q = mcΔT

10A. A 25.0 g metal sample at 98.0 °C is dropped into 100.0 g of water at 22.0 °C. The final temperature is 24.3 °C. What is the specific heat of the metal?

Solution: $q_{water} = (100.0)(4.184)(2.3) = 962$ J $= -q_{metal}$.
c = 962 / [(25.0)(98.0 − 24.3)] = 962/1843 = **0.522 J/(g·°C)**

10B. How much heat (kJ) must be removed to cool 250.0 g of water from 85.0 °C to 25.0 °C?

Solution: q = (250.0)(4.184)(−60.0) = −62,760 J = **−62.8 kJ** (62.8 kJ removed)

10C. The same amount of heat is added to 10.0 g of iron (c = 0.450 J/g·°C) and 10.0 g of aluminum (c = 0.897 J/g·°C). Which warms up more, and by roughly what factor?

Solution: ΔT = q/(mc), so the smaller c gives the bigger ΔT. **Iron** warms about 0.897/0.450 ≈ **2.0 times** as much.

Original Question 11 · Thermochemical stoichiometry

11A. $CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + 2\ H_2O(l)$ $\Delta H = -890.4$ kJ
What mass of CH_4 (16.04 g/mol) must burn to release 5.00×10^3 kJ?

Solution: mol CH_4 = 5000/890.4 = 5.615 mol × 16.04 = **90.1 g**

11B. Using ΔH = −2220 kJ for C_3H_8 combustion (same equation as the original question), how much heat is released when 15.0 L of propane at 25.0 °C and 1.00 atm is burned?

Solution: n = PV/RT = (1.00)(15.0)/[(0.08206)(298.15)] = 0.613 mol.
q = 0.613 × 2220 = **1.36×10^3 kJ released**

11C. $2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(l)$ $\Delta H = -571.6$ kJ
How much heat is released when 4.00 g of H_2 reacts with 16.0 g of O_2?

Solution: mol H_2 = 4.00/2.016 = 1.98 mol (needs 0.992 mol O_2); mol O_2 = 16.0/32.00 = 0.500 mol → **O_2 limiting**.
q = 0.500 mol O_2 × 571.6 kJ/mol O_2 = **285.8 kJ released**

Part III — Gases

Original Question 12 · Kinetic molecular theory postulates

12A. Rewrite each false statement so it is true:

(a) Gas molecules move in uniform, predictable paths.
(b) Collisions between gas molecules are perfectly inelastic.
(c) Average kinetic energy is inversely proportional to Kelvin temperature.
(d) Gas molecules attract each other strongly during collisions.

Solution: (a) Molecules move in constant, **random**, straight-line motion. (b) Collisions are perfectly **elastic** (no net KE lost). (c) KE_{avg} is **directly** proportional to Kelvin temperature. (d) Forces between molecules are **negligible**.

12B. Which postulate best explains each observation? (a) Gases are easy to compress. (b) A sealed gas at constant temperature keeps the same pressure forever instead of "running down." (c) Doubling the Kelvin temperature doubles the average kinetic energy.

Solution: (a) **Molecular volume is negligible** (mostly empty space). (b) **Collisions are elastic**. (c) **$KE_{avg} \propto$ T (Kelvin)**.

Original Question 13 · Real vs. ideal gases

13A. Rank these from most ideal to least ideal behavior for N_2: (A) 500 K, 0.5 atm (B) 200 K, 50 atm (C) 300 K, 1 atm (D) 500 K, 50 atm

Solution: **A > C > D > B**. Most ideal is high T with low P; least ideal is low T with high P.

13B. At the same T and P, which deviates more from ideal behavior: He or SO_2? Explain using a and b.

Solution: **SO_2**. It is larger (bigger b, its own volume matters more) and polar with stronger attractions (bigger a).

13C. In the van der Waals equation, what does each constant correct for? Does the a-term make the real pressure higher or lower than ideal?

Solution: **a** corrects for intermolecular attractions; **b** corrects for the volume the molecules themselves occupy. Attractions soften collisions with the wall, so the a-term makes the real pressure **lower**.

Original Question 14 · Van der Waals calculation

14A. Calculate the pressure of 2.00 mol of CO_2 in 1.50 L at 300.0 K (a) with the ideal gas law and (b) with the van der Waals equation (a = 3.592 atm·L²/mol², b = 0.04267 L/mol). (c) What is the percent difference, and which correction dominates?

Solution: (a) P = (2.00)(0.08206)(300.0)/1.50 = **32.8 atm**
(b) nRT/(V − nb) = 49.24/(1.50 − 0.0853) = 34.80 atm; a(n/V)² = 3.592(2.00/1.50)² = 6.39 atm; P = **28.4 atm**
(c) About **15%** below ideal. The real pressure is lower, so the **attraction (a) term dominates**.

Original Question 15 · KE_{avg} depends on T; speed depends on T and molar mass

15A. Container A holds 2 mol N_2 at 300 K; container B (same volume) holds 1 mol N_2 at 600 K. Compare (a) P (b) KE_{avg} (c) v_{rms}.

Solution: (a) P ∝ nT: 600 vs 600 → **equal**. (b) KE_{avg} ∝ T → **B is 2× larger**. (c) v_{rms} ∝ √T → **B is √2 ≈ 1.41× faster**.

15B. Calculate v_{rms} of O_2 at 25.0 °C (R = 8.314 J/mol·K, M in kg/mol). At what temperature would He (4.003 g/mol) have the same v_{rms}?

Solution: v = √[3(8.314)(298.15)/0.03200] = **482 m/s**.
Same speed means same T/M: T_{He} = 298.15 × (4.003/32.00) = **37.3 K**

Original Question 16 · Charles's law and combined gas law

16A. A balloon has a volume of 2.50 L at 22.0 °C. It is taken outside where it is −15.0 °C (P constant). What is its new volume?

Solution: V_2 = 2.50 × (258.15/295.15) = **2.19 L**

16B. A weather balloon holds 125 L of He at 1.00 atm and 20.0 °C. It rises to where P = 0.350 atm and T = −40.0 °C. What is its volume?

Solution: V_2 = 125 × (1.00/0.350) × (233.15/293.15) = **284 L**

16C. A student says “heating a gas from 20 °C to 40 °C doubles its volume.” Is that right? By what factor does V actually change?

Solution: **No**. Use Kelvin: 313.15/293.15 = **1.07**, about a 7% increase.

Original Question 17 · Dalton's law and mole fractions

17A. 6.00 g of Ne and 8.00 g of O_2 are placed in a 5.00 L flask at 25.0 °C. Find the total pressure and each partial pressure.

Solution: mol Ne = 6.00/20.18 = 0.297; mol O_2 = 8.00/32.00 = 0.250; total = 0.547 mol. P_{tot} = (0.547)(0.08206)(298.15)/5.00 = **2.68 atm**; P(Ne) = **1.45 atm**; P(O_2) = **1.22 atm**

17B. H_2 is collected over water at 25.0 °C. The total pressure is 755.0 mmHg and the vapor pressure of water is 23.8 mmHg. If 0.250 L of gas is collected, how many moles of H_2 are there?

Solution: P(H_2) = 755.0 − 23.8 = 731.2 mmHg = 0.962 atm.
n = (0.962)(0.250)/[(0.08206)(298.15)] = **9.83×10^{-3} mol**

17C. A flask holds equal masses of He and Ar. Which has the larger partial pressure, and by about how much?

Solution: **He**, because equal masses means more moles of the lighter gas: ratio ≈ 39.95/4.003 ≈ **10 times**.

Original Question 18 · Graham's law (find a molar mass)

18A. An unknown noble gas takes 2.24 times as long to effuse as He under the same conditions. Identify it.

Solution: $t_x/t_{He} = \sqrt{(M_x/M_{He})} \rightarrow M_x = 4.003 \times 2.24^2 = 20.1$ g/mol → **neon**

18B. An alkane C_nH_{2n+2} effuses at 0.742 times the rate of O_2. Find its molar mass and formula.

Solution: M = 32.00/(0.742)² = 58.1 g/mol. 14n + 2 = 58.1 → n = 4 → **C_4H_{10}** (butane)

18C. Uranium enrichment separates $^{235}UF_6$ (349.03 g/mol) from $^{238}UF_6$ (352.04 g/mol) by effusion. What is the ratio of their effusion rates?

Solution: $rate_{235}/rate_{238} = \sqrt{(352.04/349.03)}$ = **1.0043**. That is a tiny difference, which is why thousands of stages are needed.

Original Question 19 · Effusion rate depends on both T and M

19A. Which two pairs have essentially equal effusion rates? H_2 at 100 K, He at 200 K, CH_4 at 400 K, O_2 at 800 K

Solution: T/M: H_2 = 49.6, He = 50.0, CH_4 = 24.9, O_2 = 25.0.
H_2(100 K) ≈ He(200 K), and **CH_4(400 K) ≈ O_2(800 K)**. The first pair is faster.

19B. At what temperature would SO_2 (64.07 g/mol) effuse at the same rate as CH_4 (16.04 g/mol) at 300 K?

Solution: Equal T/M: T = 300 × (64.07/16.04) = **1.20×10^3 K**

Part IV — Entropy and Gibbs Free Energy

Original Question 20 · Predicting the sign of ΔS

20A. Predict the sign of ΔS: (a) $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$ (b) NaCl(s) dissolving in water (c) $H_2O(l)$ heated from 25 °C to 75 °C (d) $2\ H_2O_2(l) \rightarrow 2\ H_2O(l) + O_2(g)$ (e) frost forming on a car window

Solution: (a) **−** (4 mol gas → 2) (b) **+** (c) **+** (d) **+** (gas produced) (e) **−** (gas → solid)

20B. Rank by standard molar entropy, lowest to highest: (a) $H_2O(g)$, $H_2O(s)$, $H_2O(l)$ (b) $C_3H_8(g)$, $CH_4(g)$, $C_2H_6(g)$

Solution: (a) **$H_2O(s) < H_2O(l) < H_2O(g)$** (b) **$CH_4 < C_2H_6 < C_3H_8$** (more atoms give more ways to move)

Original Question 21 · ΔG = ΔH − TΔS and the crossover temperature

21A. A reaction has ΔG° = −12.0 kJ/mol at 300 K and +8.0 kJ/mol at 500 K. Assuming ΔH° and ΔS° are constant, find ΔS°, ΔH°, and the crossover temperature. Is it spontaneous above or below that temperature?

Solution: Slope = ΔΔG/ΔT = (8.0 − (−12.0))/200 = +0.100 kJ/K = −ΔS → **ΔS° = −100 J/(mol·K)**.
ΔH° = ΔG + TΔS = −12.0 + 300(−0.100) = **−42.0 kJ/mol**.
T = ΔH/ΔS = 42.0/0.100 = **420 K**; spontaneous **below 420 K**.

21B. ΔH° = +45.0 kJ/mol and ΔS° = +125 J/(mol·K). At what temperature does the reaction become spontaneous, and in which direction of temperature change?

Solution: T = 45,000/125 = **360 K**; spontaneous **above 360 K** (both ΔH and ΔS positive).

Original Question 22 · Phase transitions: T = ΔH/ΔS

22A. Ethanol boils at 78.4 °C with ΔH_{vap} = 38.6 kJ/mol. What is ΔS_{vap}?

Solution: ΔS = ΔH/T = 38,600/351.55 = **110 J/(mol·K)**

22B. For ice: ΔH_{fus} = 6.01 kJ/mol, ΔS_{fus} = 22.0 J/(mol·K). (a) Calculate the melting point. (b) Calculate ΔG for melting at −10.0 °C. Is melting or freezing spontaneous there?

Solution: (a) T = 6010/22.0 = **273 K**
(b) ΔG = 6.01 − (263.15)(0.0220) = **+0.22 kJ/mol** → melting is nonspontaneous; **freezing is spontaneous**.

Original Question 23 · Temperature dependence of spontaneity

23A. $2\ NO_2(g) \rightarrow N_2O_4(g)$ ΔH° = −57.2 kJ, ΔS° = −175.8 J/K
(a) Is it spontaneous at 25 °C? (b) At 100 °C? (c) What is the crossover T? (d) Is the slope of ΔG vs T positive or negative?

Solution: (a) ΔG = −57.2 + 298.15(0.1758) = **−4.8 kJ → spontaneous**
(b) ΔG = −57.2 + 373.15(0.1758) = **+8.4 kJ → nonspontaneous**
(c) T = 57.2/0.1758 = **325 K** (52 °C) (d) slope = −ΔS = **positive**

23B. A plot of ΔG° vs T is a straight line with a negative slope and a y-intercept of +50 kJ. What are the signs of ΔH and ΔS? Is the reaction favored at high or low T?

Solution: Intercept = ΔH → **ΔH > 0**. Slope = −ΔS < 0 → **ΔS > 0**. Spontaneous at **high T**.

Original Question 24 · The four ΔH/ΔS sign combinations

24A. Reaction C is exothermic with ΔS < 0. Reaction D is endothermic with ΔS < 0. Which is never spontaneous? When is the other one spontaneous?

Solution: **D is never spontaneous** (ΔG always > 0). **C is spontaneous only at low T.**

24B. Classify: (a) $2\ H_2O_2(l) \rightarrow 2\ H_2O(l) + O_2(g)$, $\Delta H < 0$ (b) photosynthesis, $6\ CO_2(g) + 6\ H_2O(l) \rightarrow C_6H_{12}O_6(s) + 6\ O_2(g)$, $\Delta H > 0$

Solution: (a) $\Delta H -$, $\Delta S +$ → **spontaneous at all T** (b) $\Delta H +$, $\Delta S -$ (6 mol gas → 6 mol gas plus an ordered solid; $\Delta S < 0$) → **never spontaneous on its own**, which is why light energy must drive it.

Original Question 25 · Rearranging ΔG = ΔH − TΔS

25A. For $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$ at 25.00 °C, $\Delta G° = -33.0$ kJ and $\Delta S° = -198.7$ J/K. Find $\Delta H°$.

Solution: $\Delta H = \Delta G + T\Delta S = -33.0 + (298.15)(-0.1987) =$ **−92.2 kJ**

25B. A reaction has $\Delta H° = -120.0$ kJ and $\Delta S° = -150.0$ J/K. At what temperature is $\Delta G° = -60.0$ kJ?

Solution: $-60.0 = -120.0 - T(-0.1500) \rightarrow T = 60.0/0.1500 =$ **400. K**

Original Question 26 · Full thermodynamic analysis from tables

26A. $2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g)$

Data: SO_2: $\Delta H°_f = -296.8$ kJ/mol, $S° = 248.2$ J/(mol·K); O_2: $S° = 205.0$; SO_3: $\Delta H°_f = -395.7$, $S° = 256.8$

(a) $\Delta H°$ (b) $\Delta S°$ (c) $\Delta G°$ at 298 K, and is it spontaneous? (d) Above what temperature does it become nonspontaneous?

Solution: (a) $2(-395.7) - 2(-296.8) =$ **−197.8 kJ**

(b) $2(256.8) - [2(248.2) + 205.0] =$ **−187.8 J/K**

(c) $-197.8 - 298(-0.1878) =$ **−141.8 kJ → spontaneous**

(d) $T = 197.8/0.1878 =$ **1.05×10^3 K**

26B. For $CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + 2\ H_2O(l)$, $\Delta S° = -242.8$ J/K. Given $S°$: $CO_2 = 213.7$, $H_2O(l) = 69.9$, $O_2 = 205.0$ J/(mol·K), find $S°$ of $CH_4(g)$.

Solution: $\Delta S = [213.7 + 2(69.9)] - [S°(CH_4) + 2(205.0)] \rightarrow -242.8 = 353.5 - S° - 410.0 \rightarrow S°(CH_4) =$ **186.3 J/(mol·K)**

Part V — Light and Photons

Original Question 27 · Visible spectrum: wavelength, frequency, energy

27A. Which visible color has (a) the shortest wavelength (b) the lowest frequency (c) the lowest energy per photon?

Solution: (a) **violet** (b) **red** (c) **red**

27B. Light A is 650 nm and light B is 450 nm. Which is roughly red and which is blue? Which has the higher frequency?

Solution: A (650 nm) ≈ **red**, B (450 nm) ≈ **blue**. **B** has the higher frequency and energy.

27C. Rank by increasing photon energy: X-rays, microwaves, visible, radio, ultraviolet, gamma rays, infrared.

Solution: **radio < microwave < infrared < visible < ultraviolet < X-ray < gamma**

Original Question 28 · What controls photon energy

28A. Select all that INCREASE the energy of a single photon: (a) increase wavelength (b) increase frequency (c) increase intensity (d) decrease wavelength (e) shift from red light to blue light

Solution: **(b), (d), (e)**. Intensity changes only the number of photons.

28B. A dim violet light and a very bright red light shine on a surface. Which has more energetic photons? Which delivers more photons per second?

Solution: **Violet** has more energy per photon. The **bright red** light delivers more photons per second (brightness = number of photons).

Original Question 29 · Energy per mole of photons

29A. What is the energy in kJ/mol of photons with λ = 532 nm (green laser)?

Solution: $E = hc/\lambda = (6.626 \times 10^{-34})(2.998 \times 10^{8})/(532 \times 10^{-9}) = 3.734 \times 10^{-19}$ J. $\times\ 6.022 \times 10^{23} = 2.249 \times 10^{5}$ J/mol = **225 kJ/mol**

29B. A photon has an energy of 2.50 eV. What is its wavelength in nm?

Solution: $E = 2.50 \times 1.602 \times 10^{-19} = 4.005 \times 10^{-19}$ J. $\lambda = hc/E = 4.96 \times 10^{-7}$ m = **496 nm**

29C. How many photons are in a 1.00 mJ pulse of 405 nm laser light?

Solution: $E_{photon} = hc/\lambda = 4.905 \times 10^{-19}$ J. $n = 1.00 \times 10^{-3}/4.905 \times 10^{-19}$ = **2.04×10^{15} photons**

Original Question 30 · Wavelength from energy: λ = hc/E

30A. The C–C bond energy is 348 kJ/mol. What is the longest wavelength (nm) of light that can break one C–C bond? What region of the spectrum is this?

Solution: Per bond: $348{,}000/6.022 \times 10^{23} = 5.779 \times 10^{-19}$ J. $\lambda = hc/E = 3.44 \times 10^{-7}$ m = **344 nm**, in the **ultraviolet**.

30B. An FM radio station broadcasts at 101.1 MHz. Find the wavelength (m) and the energy of one photon (J).

Solution: $\lambda = c/\nu = 2.998 \times 10^{8}/1.011 \times 10^{8}$ = **2.97 m**. $E = h\nu = (6.626 \times 10^{-34})(1.011 \times 10^{8})$ = **6.70×10^{-26} J**